AEGIS-RF: Supporting Documentation Package

Icicle Kit Board Audit, Criteria Traceability Matrix, Technical Risk Register, Evaluation Measurement Protocol, and Verification Checklist

1. Microchip Icicle Kit Board-Document Audit

This audit translates the five official Microchip board documents into an empirically grounded hardware implementation plan for Track 2 (PolarFire SoC Icicle Kit):

Document Title	Official Doc ID & Rev	Technical Grounding & Usage in AEGIS-RF
PolarFire SoC Icicle Kit User Guide	DS60001679E (2026)	Board identity, memory interface, I/O banks, expansion connectors, PAC1934 channels, LEDs/switches, 50 MHz fabric oscillator.
PolarFire SoC Icicle Kit QuickStart Guide	DS50003930A (2025)	254K logic elements, board power/USB-UART bring-up, Linux serial access, and jumper baseline.
PolarFire SoC Icicle Kit Brochure	DS00003611A (2020)	Five-core RISC-V MSS summary, 2 GB LPDDR4 x32, SPI flash, eMMC/SD, dual Gigabit Ethernet, and PCIe.
Icicle Kit Rev. 1.0 Schematics	DVP-102-000536-001 (2020)	Actual board wiring, bank assignments, LPDDR4, clock distribution, PAC1934 power sensing, and peripheral pinouts.
PolarFire SoC FPGA Board Design Guidelines	DS60001681H (2026)	Power integrity, I/O bank voltages, clock/reset, DDR and high-speed interface constraints.

1.1 Key Architectural Decisions Carried Into Proposal

- **Fixed FFT Fabric Pipeline:** Fabric executes deterministic fast-loop DSP: fixed 2048-point CoreFFT, spectral magnitude, continuous full-band candidate scan, and windowed CA-CFAR.
- **RISC-V Slow-Loop Control:** RISC-V Linux MSS executes supervisory control: confidence scoring, hysteresis FSM, run-time mode selection, and Ethernet telemetry logging.
- **Reference Design Validation:** Pre-custom baseline bring-up: boot vendor reference Linux image and verify official AXI4 streaming DMA demo before inserting custom DSP.
- **Telemetry Boundaries:** PAC1934 rail telemetry: measures VDD (Core), VDD25, VDDA25, and VDDA rails via MSS I2C; external calibrated power meter used for total wall power.
- **Parameter Switching:** Zero DPR dependency: dynamic adaptation achieved via frame-synchronous shadow register latching (AXI-Lite 0x00-0x1C), eliminating bitstream reload.

2. Official Contest Criteria Traceability Matrix

The following matrix maps the AEGIS-RF proposal features directly to both Proposal-Stage (100%) and Final-Stage (100%) judging rubrics:

Rubric & Weight	Key Evaluation Criteria	AEGIS-RF Technical Feature	Verification Evidence
Proposal: Technical Feasibility (35%)	Realistic architecture, vendor IP utilization, timing closure, resource margins.	Option B Fixed 2048 CoreFFT, ping-pong LPDDR4 DMA, verified MSS Linux stack.	Libero SoC 2024.1 synthesis reports, timing closure at >= 100 MHz, 37.8% LE utilization.
Proposal: Innovation & Creativity (25%)	Originality, edge adaptation, novelty beyond static spectrum analyzers.	Confidence-Aware Runtime Mode Switching; continuous full-band candidate awareness.	Multi-feature confidence metric (PNR, persistence, margin); frame-synchronous latching.
Proposal: Problem Statement & Clarity (20%)	Clear edge radio challenges, rigorous methodology, SWaP-C trade-offs.	Tradeoff study: energy consumption vs RF detection fidelity under non-stationary traffic.	Clean 21-section proposal, 5 professional figures, explicit 3-way evaluation protocol.
Proposal: Potential Impact (20%)	Real-world applicability, tactical UAV, cognitive radio, smart spectrum.	Dynamic SWaP-C reduction for edge sensor nodes; autonomous quiescent scaling.	PAC1934 multi-rail power telemetry; energy-per-sample (µJ) metric across test scenarios.
Final: Code & Implementation (35%)	RTL code quality, timing closure, driver robustness, zero sample loss.	Clean modular Verilog RTL, CoreFFT AXI-Stream, UIO Linux driver, zero drop ping-pong.	Bit-accurate RTL-to-MATLAB testbench logs, continuous streaming throughput verification.
Final: Video Demonstration (20%)	Clear 3-5 min video showing live working Icicle Kit hardware.	Split-screen video: Icicle Kit board LEDs/UART, live PC dashboard, PAC1934 telemetry.	Live demonstration of dynamic mode switching in response to synthetic burst interference.
Final: Project Documentation (20%)	Concise 10-page final report, README build guide, open reproducibility.	Pre-structured 10-page report template, step-by-step Libero/Linux build README.	Complete GitHub repository with reproducible MATLAB test harness and seed scripts.

3. Comprehensive Technical Risk Register

Summary of the top technical risks, severity ratings, pre-planned mitigations, and validated fallback architectures:

Risk ID & Item	Sev / Prob	Technical Impact	Pre-Planned Mitigation Strategy	Fallback Architecture
R-01: CoreFFT Licensing / Configuration	High / Med	CoreFFT IP not available or transform unsupported.	Verify CoreFFT license early; pin v7.1 / v8.0 streaming pipeline configuration.	Implement Radix-2 / Radix-4 pipelined streaming FFT in open Verilog.
R-02: LPDDR4 DMA Bandwidth Saturation	Med / Low	DMA throughput bottleneck causes dropped FFT frames.	Ping-pong circular buffers; compact 16-bit packed spectral descriptors; IRQ throttling.	Store raw FFT in fabric LSRAM; transfer only CFAR candidate peak descriptors to CPU.
R-03: PAC1934 I2C Telemetry Latency	Med / Low	I2C read latency limits real-time energy logging.	Decouple power monitoring loop (10 Hz periodic timer in Linux userspace).	Log average power over 1-second epochs; correlate with active mode state.
R-04: Mode Switching Frame Corruption	High / Low	Mid-frame parameter changes corrupt active CFAR detection.	Implement shadow registers; latch new parameters strictly at sof_sync frame boundary.	Defer register updates until DMA buffer completion interrupt.
R-05: Timing Closure at Target Clock	Med / Low	Fabric DSP pipeline fails timing closure at 125 MHz.	Register all arithmetic stages; pipelined multipliers; use dedicated MACC blocks.	Scale fabric operating frequency to 100 MHz (supported by 50 MHz oscillator PLL).

4. Evaluation & Measurement Protocol

The project enforces a strict, reproducible three-way comparative benchmark protocol across identical held-out RF test datasets:

Configuration	Architecture Description	FFT Transform	Downstream Workload	Primary Purpose
Config A: Baseline Fixed-Coarse	Static low-power baseline; minimal feature analysis.	Fixed 2048 CoreFFT	Static 256-bin window, K_CFAR=4.5, moments disabled.	Establishes minimum dynamic power baseline.
Config B: Baseline Fixed-Detailed	Static high-fidelity baseline; maximal feature analysis.	Fixed 2048 CoreFFT	Static 2048-bin window, K_CFAR=2.5, full moments enabled.	Establishes maximum detection fidelity baseline.
Config C: AEGIS-RF Adaptive	Confidence-aware dynamic workload scaling.	Fixed 2048 CoreFFT	Dynamically scaled (256/1024/2048 bins) based on C_det.	Evaluates dynamic energy savings while preserving detection.

4.1 Predeclared Measurement Boundaries

- **Input Test Vector Parity:** All three configurations process the identical MATLAB-synthesized 10,000-frame complex I/Q test vector dataset.
- **CFAR Operating Point Parity:** Probability of False Alarm (Pfa) calibrated to 10^-3 across all configurations before comparing Probability of Detection (Pd).
- **Power Telemetry Boundary:** Power measured via onboard PAC1934 telemetry across VDD (core 1.0V), VDD25, VDDA25, and VDDA rails, validated against external USB meter.
- **Energy Metric Standard:** Energy-per-sample (µJ) calculated as: E_sample = (P_active * T_frame) / N_samples.